Florence, Italy 12-16 July 2015
ICSV22
The 22nd International Congress on Sound and Vibration

ACOUSTIC PROPERTIES OF STONE AGGREGATES USED IN COOL ROOFS

Francesco Asdrubali, Franco Cotana, Anna Laura Pisello and Nicholas Mencarelli

University of Perugia, Department of Engineering, Via G. Duranti 63, 06125 Perugia, Italy
e-mail: francesco.asdrubali@unipg.it

Francesco D'Alessandro

University of Perugia, Department of Civil and Environmental Engineering, Via G. Duranti 63, 06125 Perugia, Italy

Samuele Schiavoni

CIRIAF – Inter University Research Centre for Environment and Pollution "Mauro Felli", Via G. Duranti 67, 06125 Perugia, Italy

In the last couple of decades cool roofs, i.e. roof surfaces exposed to solar radiation characterized by high solar reflectance (albedo) and thermal emittance, are being intensively studied for their ability to reduce the roof temperature, which leads to a lower heat transfer into the building. This acknowledged phenomenon produces better indoor thermal comfort conditions and reduced energy requirement for cooling, resulting in lower greenhouse gas emissions. Additionally, cool roofs are also studied and optimized for their larger scale effect as Urban Heat Island mitigation strategy in dense cities. In this perspective, several reflective coatings and materials were used for cool roof applications, e.g. light color paintings, roof tiles and waterproof membranes. Recent research developments were also focused on the analysis of natural and sustainable materials having these properties. In particular, stone aggregates were studied for their intrinsic cooling potential with varying grain size and stone typology. Roof covered by stone aggregates can be considered as a rigid frame porous material whose acoustic properties can lead to similar advantages to the thermal ones: (i) a small scale benefit, i.e. an increased sound reduction index of the roof with a consequent lower sound transfer into the building; (ii) a larger scale benefit, i.e. a reduction of the environmental noise thanks to the increased sound absorption coefficient. This paper reports the results of the measurement campaigns of sound absorption and transmission loss carried out on different samples of loose gravel with different grain size and thickness. To this aim, several experimental analyses by means of impedance tube according to the transfer function method described in ISO 10534-2 were carried out. Several comparative considerations between acoustic and thermal-energy experimental results represent the object of this contribution, with the final purpose to optimize this roof typology for its twofold potential.

1. Introduction

Building energy efficiency and indoor microclimate comfort in constructions have represented key research and innovation issues [1] during the last decades. One of the main trends in this field consisted of the analysis and the optimization of sustainable materials responsible of multiple benefits such as (i) optimization of indoor thermal comfort, (ii) improvement of acoustic performance of buildings envelope [2], (iii) positive effects of outdoor local microclimate [3]. This latter aspect was mainly focused on the proposal of new techniques aimed at reducing local climate changes phenomena such as urban heat island. This phenomenon mainly consists of the increase of urban temperatures during the course of the year, with respect to the colder suburban and green surrounding areas, less affected by anthropic activities. An important role in increasing such phenomenon is played by greenhouse gases emissions and energy consumption in buildings [4], that is the reason why many research and industrial efforts were dedicated to the proposal of new and sustainable materials and techniques [5] with a twofold aim, i.e. to reduce building energy requirement for cooling and to improve urban livability. This combined effect was shown to be produced by the so called "cool materials" that are those materials able to reflect solar radiation and to emit thermal energy much more than classic construction and paving materials. Therefore, cool materials are characterized by high solar reflectance, e.g. more than 40-50%, and high thermal emittance, e.g. more than 80% [6]. Being able to reflect solar radiation and to emit thermal energy, they are able to stay cooler than classic materials, e.g. dark bitumen asphalt, etc., and to produce a non-negligible passive cooling effect in buildings and in urban areas, when applied in outdoor paving systems exposed to solar radiation. For these acknowledged reasons, the passive cooling potential of cool materials was investigated through numerical dynamic simulation and experimental activities in several climate conditions around the world in summer and winter conditions [7-8]. Additionally, another research issue consisted in the proposal and improvement of reliable techniques and procedures to perform experimental measurements of the optic-radiative material properties, i.e. albedo/solar reflectance [9-10]. These latter efforts were also aimed at characterizing from the optical and thermal-energy point of view the performance of natural aggregated layers typically used all around the world for roof covering and urban paving. After acknowledging a non-negligible natural "cool" behavior of selected local stone in previous works by the same authors [10], this paper concerns the acoustic analysis of the performance of such high reflective gravel in the form of aggregates.

The stone aggregates layers investigated in the present research can be considered as loose granular materials, in particular as rigid frame porous media with open porosity due to the presence of interconnected pores between the particles. The complexity of the inner structure of granular materials, i.e. pores of variable sizes and shapes, introduces a higher level of difficulty in modelling their sound absorption characteristics if compared with fibrous materials, which can be studied with a number of reliable models [11]. Voronina *et al* have thoroughly investigated the sound absorption properties of loose granular materials, defining a model that takes into account the following parameters: porosity, tortuosity, specific density and characteristic particle dimension of the loose granular mix [12-13]. This model proved to be reliable if compared with experimental data obtained testing several granular materials, including vermiculite, rubber crumbs, perlite and keramsite. A large set of experimental data related to rubber crumbs obtained by end of life tires can be found in [14] while data on the sound absorption properties of loose expanded clay are reported in [15]. However the authors are aware of very few scientific papers reporting experimental data of the sound absorption properties of stone aggregates [16], while no data concerning transmission loss have been found in the literature. This latter aspect can be of interest if the material is used for cool roofs applications, since it can bring to an increase in the sound insulation of the roof.

The acoustic characterization is performed by combining the findings of thermal-energy previous studies with the results of acoustic in-lab measurement campaign carried out in this paper. In particular, several thicknesses of the gravel layer and grain sizes are tested in order to be able to compare

albedo (solar reflectance) capability with acoustic potential, with the final purpose to identify an integrated solution able to reduce building energy need for cooling, to cool urban areas and to improve the acoustic indoor-outdoor environment.

2. Materials and methods

2.1 Characteristics of the stone

The investigated materials are low-cost stones from central Italy made almost at 100% of calcium carbonate, typically used as aggregate for concrete based materials. The use of such stone as roof covering is an original aspect of this research. Given the natural high thermal emissivity of all the construction materials with no metal content and the stone light color, a reasonable good natural behavior was assumed and studied in previous works by the same authors [10]. Four already commercially available stone sizes has been chosen for the present analysis in order to integrate the optic-thermal analysis carried out in [10] with the acoustic characterization which is the main object of this work. In particular, Table 1 and Fig. 1 report grain sizes and pictures of the stones used for building roof application. The grain sizes 3 and 4 are the most common sizes usually applied over roof or external paved areas. Specific grain sizes were shown to affect the aggregate albedo which increases from about 30% to about 50% with decreasing of grain size [10], as in-field measured by the authors. The selected stone type was chosen also for being a local sustainable solution, with natural cool properties, since the albedo of generic gravels tested by the authors or other researchers never exceeds 30% [9]. Being a local material, it also has the advantage of a reduced embodied energy, due to the low energy consumption for transportation.

Table 1. Grain sizes of the tested stone.

Aggregate code	Grain size
1_xl	8–22.4 mm
2_l	4–12.5 mm
3_m	2–5.6 mm
4_s	0–4 mm



Figure 1. Pictures of the tested stones (codes are reported in Table 1).

2.2 Description of the methodology

Normal incidence absorption coefficient and transmission loss of the investigated samples were measured using an impedance tube Brüel & Kjær, model 4260, connected to a signal analyser Brüel & Kjær Pulse.

2.2.1 Normal incidence absorption coefficient

The absorption coefficient was measured according to the procedures defined by the standard ISO 10534-2 [17]. The tube was mounted in vertical position and the stones were poured into the sample holder trying to obtain an upper surface of the sample as much flat as possible (Fig. 2).



Figure 2. Top view of the samples tested in the large (d=100 mm) impedance to tube. From left to right: size 2_l, size 3_m, size 4_f.

Two configurations of the tube were used: (i) the large tube, with a diameter of 100 mm in order to investigate the frequency range of 50-1600 Hz and (ii) the small tube, with a diameter of 29 mm to cover the frequency range of 500-6400 Hz. Then the results obtained by the two tubes for a sample were combined in order to obtain a single trend in the frequency range 50-6400 Hz.

The first available size, code 1_xl, was discarded since the voids created by the grains were too big to achieve an adequate energy loss; moreover the size of the single grain was of the same order of magnitude with respect to the small impedance tube. The remaining sizes, namely 2_l, 3_m and 4_s, were tested with four different values of thickness: 10 mm, 20 mm, 40 mm and 80 mm. The first three values of thickness can be used when the stones are placed on a roof to improve the solar reflectance or, more commonly, to protect the watertight sheath from the sun, wind, rain and other meteorological causes of degradation. The 80 mm thickness was investigated for characterizing the stones laid in external paved areas.

Each configuration was tested three times and the results averaged in order to make the results less sensitive to the heterogeneity of the studied material.

2.2.2 Transmission loss

The transmission loss was measured by means of the 'two-load' transfer function method [18-19], which is an extension of the transfer function method used for the normal incidence sound absorption measurements [17]. This method requires a four microphone impedance tube and four sets of measurements: two without the sample under test with a rigid and an anechoic termination and two in the same configurations with the sample under test. This method allows to evaluate the complex amplitudes of the direct, reflected and propagated plane wave components in the upstream and downstream tube sections. These components are correlated by parameters depending on the acoustic properties of the sample and one of them is the transmission coefficient τ, which is needed to calculate the transmission loss TL as follows:

$$TL = 10\log_{10}\frac{1}{|\tau|^2} \text{ [dB]} \tag{1}$$

As in the previous case, the impedance tube was mounted vertically for transmission loss tests too. The sample holder of the impedance tube was specifically modified in order to include the loose stones in vertical positions. In particular a piece of a PVC tube was cut and a face covered with a nylon mesh (Fig. 3, on the left); the exterior diameter of the tube was exactly 100 mm so it fitted perfectly in the sample holder without falling (Fig. 3 on the right). Preliminary tests were performed on the modified sample holder without stones with the aim of evaluating its influence on the measurement results: Fig. 3 shows that the mesh had a negligible effect on the acoustic properties of the samples, except from some peaks of low absolute amplitude in the frequency range 200-300 Hz; above 300 Hz, transmission loss is constantly very close to 0 dB.

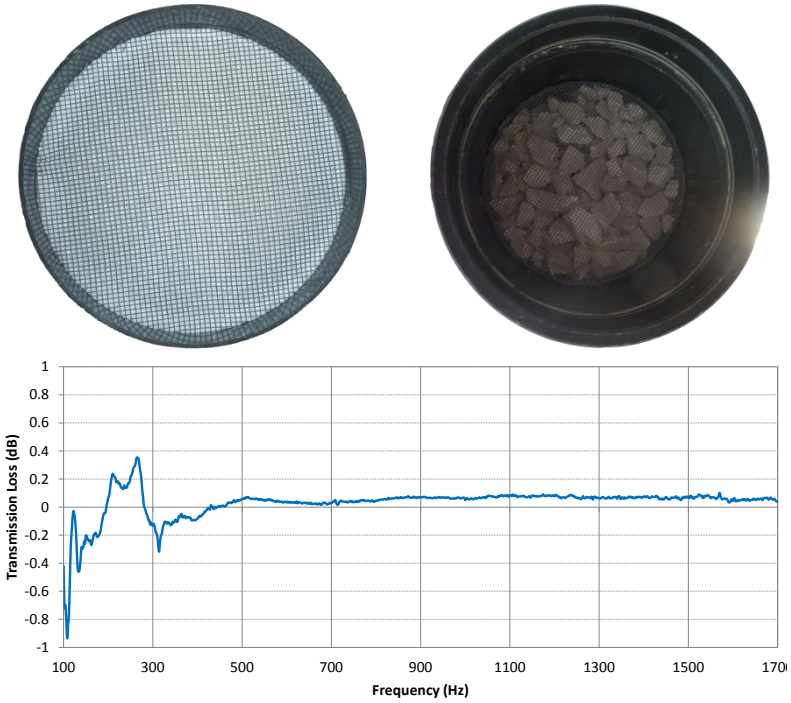


Figure 3. View of the sample holder with the nylon mesh used to contain the stones during the transmission loss tests and trend of the transmission loss of the mesh alone.

Only two sizes of stones were tested as far as transmission loss: size 2_l and 3_m. Size 1_xl was discarded as for the sound absorption tests while size 4_s was discarded because it showed the worst performance of sound absorption (see following section) and also because the mesh was not able to contain the thinnest component of the sample (see table 1 for grain sizes). Three thicknesses were tested for each grain size: 20 mm, 40 mm and 80 mm; each test was repeated three times and the results averaged. Transmission loss tests were performed employing the large tube configuration only, so the results are significant in the frequency range 100-1600 Hz.

3. Results

3.1.1 Normal incidence absorption coefficient

Fig. 4-6 report the trends of the normal incidence sound absorption coefficient in the frequency range 100-6400 Hz for the three investigated grain sizes (2_l, 3_m and 4_s) and for four different values of thickness (10 mm, 20 mm, 40 mm and 80 mm).

As expected, for the same grain size increasing values of the thickness lead to higher values of the sound absorption coefficient and to a shift of the first peak of sound absorption towards lower frequencies, as demonstrated in Fig. 4 for size 2_l and in Fig. 5 for size 3_m. The first absorption peaks for 10 mm thick samples fall outside the investigated frequency range. Furthermore for the same thickness the sound absorption was found to be higher for smaller grain sizes, according to previous researches by the authors [14-15].

However these statements were not verified for the grain size 4_s: Fig. 6 shows that in this case the expected trends are not obtained and the absorption coefficient keeps at a quite constant and low value even changing the values of the thickness. This is mainly due to the fact that the smallest fraction of the 4_s grain size filled the open pores created by the largest fraction and obstructed the air flow inside the voids (see Fig. 2). Therefore the absorption mechanisms were due to the top layers of the sample only and small changes of sound absorption were detected when the thickness increased. As far as the sound absorption, the best performing grain size is the one named 3_m.

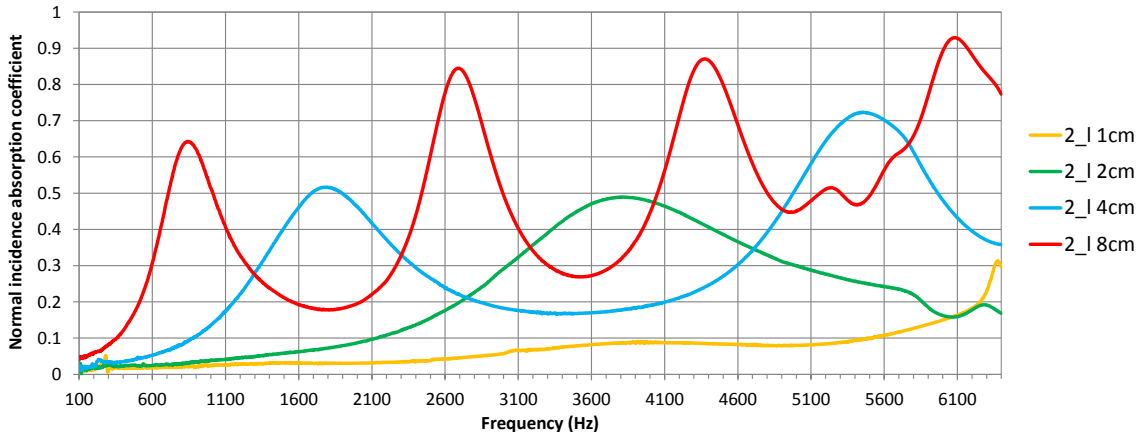


Figure 4. Normal incidence sound absorption coefficient of the grain size 2_l.

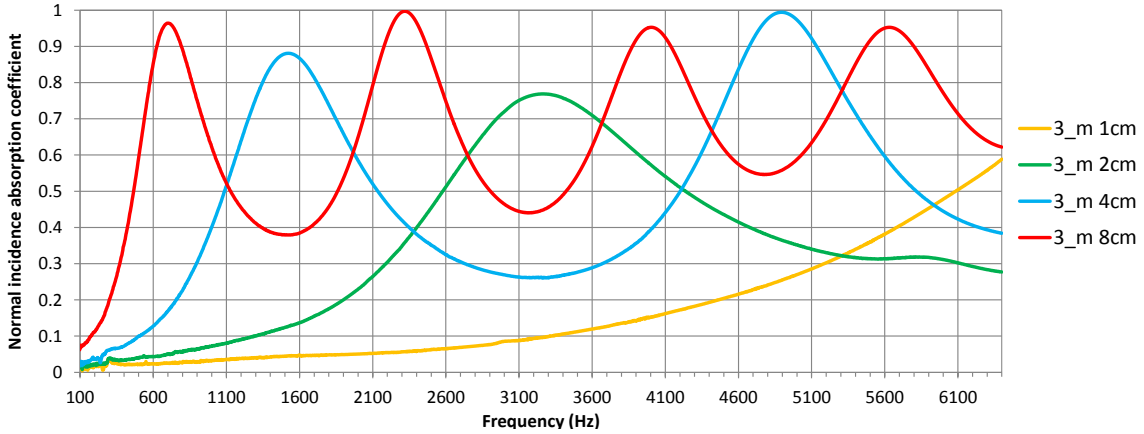


Figure 5. Normal incidence sound absorption coefficient of the grain size 3_m.

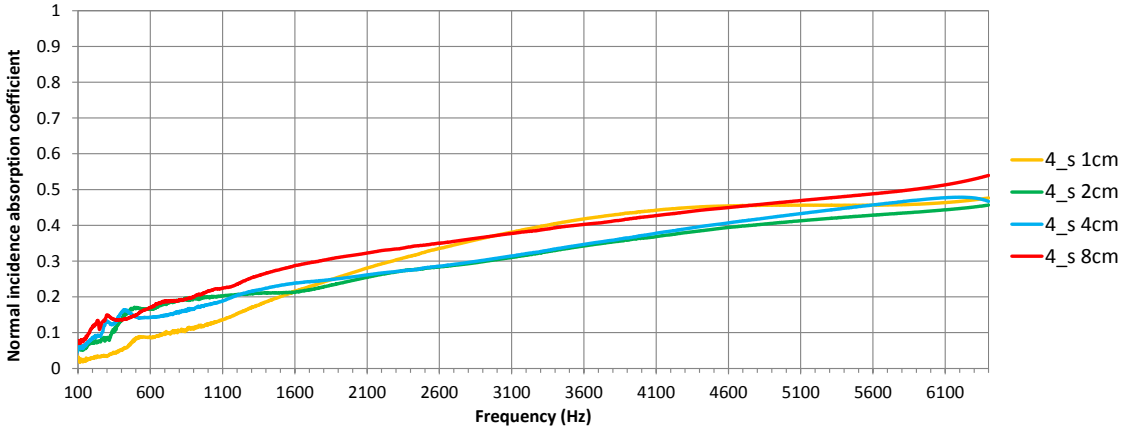


Figure 6. Normal incidence sound absorption coefficient of the grain size 4_s.

3.1.2 Transmission loss

Fig. 7 reports the trends of the transmission loss in the frequency range 100-1600 Hz for the two investigated grain sizes (2_l and 3_m) for different values of thickness (20 mm, 40 mm and 80 mm).

The results show that for each investigated thickness the measured transmission loss of the grain size 3_m is always 1-2 dB higher than the one measured for the grain size 4_l. As expected, transmission loss increases with increasing values of the thickness, reaching a maximum value of 7 dB at 700 Hz for a 80 mm sample made of stones with grain size 3_m. The influence of the mesh used for containing the stones inside the sample holder can be noticed around 300 Hz.

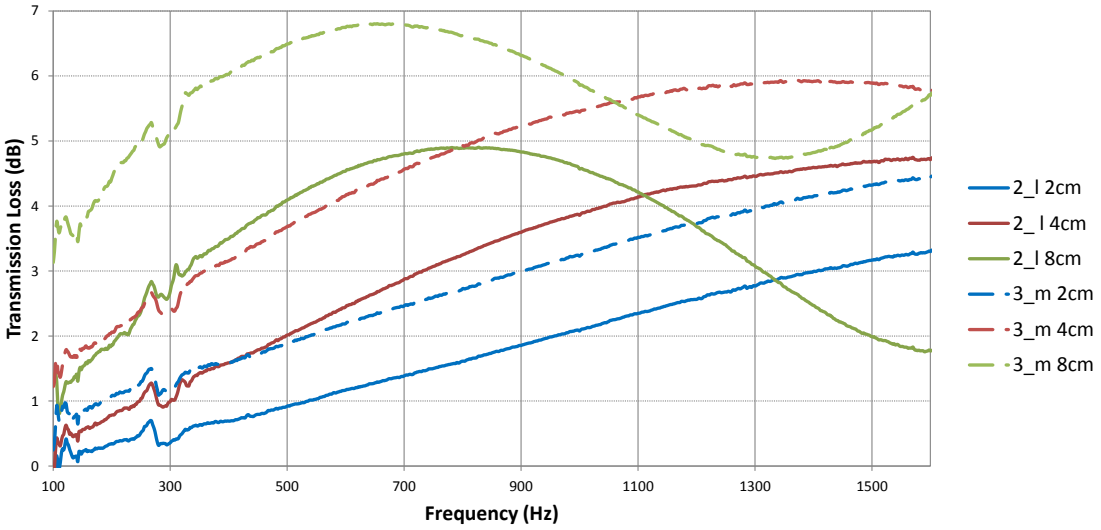


Figure 7. Transmission loss of the grain size 2_l and 3_m.

4. Conclusions

The paper reports the results of an experimental campaign carried out at the Acoustics Lab of the University of Perugia aimed at evaluating the sound absorption and the transmission loss of various samples of local natural stones used by the authors [10] for cool roofs applications. Among the tested grain sizes, the one showing the best acoustic performance, both for absorption and insulation, is the one made of grains with diameters ranging from 2 mm to 5.6 mm: a 40 mm layer of these stones showed to achieve relatively good values of sound absorption (0.5-0.9) in the frequency range of 1000-1500 Hz. In the same range the transmission loss was about 5-6 dB. This frequency range is important since the normalized road traffic spectrum defined by the standard EN 1793-3 [20] is centered at 1000 Hz. Thicker layers would bring to higher values of sound absorption and transmission loss but the weight would limit their use for cool roofs, whereas they could be considered for installation on grounds, such as external paved areas.

Taking into account the results achieved by the authors in previous studies focused on the passive cooling potential of the selected natural aggregates, this work was built upon the thermal-energy and environmental performance of the proposed solution: the results show that the cool gravel could be suitable also for acoustic purpose. In particular the acoustic benefits produced by the stone aggregates are very well documented for the relatively fine grain size (3_m), according to the albedo field measurements of the gravels, showing a relatively good solar reflectance values. In fact, the small size of stones was characterized by about 45% value of albedo, i.e. more than 30% more than a classic bitumen based membrane for roof application.

Finally, this research confirmed the promising potential of the sustainable, low cost, and local solution as roof covering and urban paving from a both acoustic and thermal-energy point of view.

REFERENCES

1 Peruzzi, L., Salata, F., De Lieto Vollaro, A. and De Lieto Vollaro, R. The reliability of technological systems with high energy efficiency in residential buildings, *Energy and Buildings*, **68** (PART A), 19-24, (2014).

2 Asdrubali, F. and D'Alessandro, F. Impact sound insulation and viscoelastic properties of resilient materials made from recycled tyre granules, *International Journal of Acoustics and Vibrations*, **16** (3), 119-125, (2011).

3 Santamouris, M. Regulating the damaged thermostat of the cities - Status, impacts and mitigation challenges, *Energy and Buildings*, **91**, 43-56, (2015).

4 Buonomano, A. and Palombo, A. Building energy performance analysis by an in-house developed dynamic simulation code: An investigation for different case studies, *Applied Energy*, **113**, 788-807, (2014)

5 Pisello, A. L. and Asdrubali, F. Human-based energy retrofits in residential buildings: A cost-effective alternative to traditional physical strategies, *Applied Energy*, **133**, 224-235, (2014).

6 Kolokotroni, M. and Kolokotsa, D. Energy and environmental aspects of cool materials. *Advances in the Development of Cool Materials for the Built Environment*, Bentham Science Publishers Ltd., 231-272, (2013).

7 Synnefa, A., Santamouris, M. and Akbari, H. Estimating the effect of using cool coatings on energy loads and thermal comfort in residential buildings in various climatic conditions. *Energy and Buildings*, **39** (11), 1167-1174, (2007).

8 Pisello, A. L., Rossi, F. and Cotana, F. Summer and winter effect of innovative cool roof tiles on the dynamic thermal behavior of buildings, *Energies*, **7** (4), 2343-2361, (2014).

9 Levinson, R., Chen, S., Berdahl, P., Rosado, P. and Medina, L.A. Reflectometer measurement of roofing aggregate albedo, *Energy*, **100**, 159-171, (2014).

10 Pisello, A. L., Pignatta, G., Castaldo, V. L. and Cotana, F. Experimental analysis of natural gravel covering as cool roofing and cool pavement, *Sustainability (Switzerland)*, **6** (8), 4706-4722, (2014).

11 Allard, J. F. and Atalla, N. *Propagation of Sound in Porous Media*, John Wiley & Sons, United Kingdom, (2009).

12 Voronina, N. N. and Horoshenkov, K. V. A new empirical model for the acoustic properties of loose granular media, *Applied Acoustics*, **64**, 415-432, (2003).

13 Voronina, N. N. and Horoshenkov, K. V. Acoustic properties of unconsolidated granular mixes, *Applied Acoustics*, **65**, 673-691, (2004).

14 Asdrubali, F., D'Alessandro, F. and Schiavoni, S. Sound absorbing properties of materials made of rubber crumbs, *Proceedings of the 7th European Conference on Noise Control EURONOISE 2008*, Paris, France, 29 June–04 July, (2008).

15 Asdrubali, F. and Horoshenkov, K. V. The acoustic properties of expanded clay granulates, *Building Acoustics*, **9** (2), 85-98, (2002).

16 Iannace, G., Ianniello, C., Maffei, L. and Romano, R. Characteristic impedance and complex wave –number of limestone chips, *Proceedings of the 4th European Conference on Noise Control EURONOISE 2001*, Patras, Greece, 14–17 January, (2001).

17 ISO 10534-2. Acoustics – determination of sound absorption coefficient and impedance in impedance tubes – Part 2: Transfer-function method; (1998).

18 Young, J. C. and Crocker, M. J. Prediction of transmission loss in mufflers by the finite-element method, *J. Acoust. Soc. Am.*, **57**, 144–148, (1975).

19 Pispola, G., Horoshenkov, K. V. and Asdrubali, F. Transmission loss measurement of consolidated granular media, *J. Acoust. Soc. Am.*, **117** (5), 2716–2719, (2005).

20 EN 1793-3. Road traffic noise reducing devices - Test method for determining the acoustic performance - Part 3: Normalized traffic noise spectrum (1997).